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Chapter 1

Balanced Homodyne for LIGO A+

There doesn't need to be a discussion about what BHD is, but it is worth briefly mentioning what it is and the other readout methods again Context for BHD upgrade being part of A+.

1.1 BHD versus DC readout for A+

- G2000962, DDR CDR presentation A.
- G2001026, DDR CDR presentation B.
- Ken's notes on BHD benefits compared to DC readout, email on 25/06/2020 11:26
- LIGO-L2000200 Q and A. Useful technical points and things people will find confusing/may ask about.

Main benefit of BHD is reduction of SRC length noise coupling to AS port. This noise coupling is described in [1], see equation 13. This paper has some coupled

cavity maths in it. If this needs explained, look up Transfer Functions for Fields in a 3-Mirror Nested Cavity, [2]. There is an opto-mechanical coupling between the length of the SRC and arm cavity (due to radiation pressure). Reducing the DC offset reduces the radiation pressure, which reduces this coupling.

estimate reduction of SRC length noise coupling Equation for SRC length noise to AS signal is (I couldn't get this to plot, need to debug this)

$$\frac{S_{\text{as}}}{S_{\text{dc}}} = 4g_p^2 g_s^2 r_a^2 \epsilon \left(\frac{\partial L_-}{\partial l_s} \right) \frac{k \Delta l_s}{1 + s_{\text{rse}}} \quad (1.1)$$

the TEM00 is decreased to fringe offset (rms + DC) + 1mW of common mode carrier, currently there is 20mA of photo-current. (shelia dwyer answer in email FW: BHD review committee request & document PREVIEW).

1.2 Derivation of path length stability

- Hang Yu BHD requirements

Analytic expression [1] and finesse simulation.

- 1.3 WS visualisations as a tool for understanding mode matching problems
- 1.4 Mode matching with astigmatism
- 1.5 Algorithm for calculating optimal 3 lens mode matching solution
- 1.6 HAM6 layout
- 1.7 Thermal lensing
- 1.8 Arm cavity power linked to AWC
- 1.9 Conclusion about BHD for A+

Bibliography

- [1] Kiwamu Izumi and Daniel Sigg. Advanced LIGO: length sensing and control in a dual recycled interferometric gravitational wave antenna. *Classical and Quantum Gravity*, 34(1):015001, dec 2016.
- [2] Malik Rakhmanov. Transfer functions for fields in a 3-mirror nested cavity. In *AIP Conference Proceedings*. AIP, 2000.